Renormalization Group Flows of Spectral Dimensions on Hierarchical Lattices: Critical Percolation, Resistance Exponents, and Transport Universality

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Abstract

A recurring idea in discrete approaches to quantum gravity and in network models of growth is that coarse-graining drives a graph *spontaneously* to a stable non-integer spectral dimension. We test this idea on a model in which the renormalization group (RG) is exact: bond percolation on the diamond hierarchical lattice (DHL).

The edge-retention probability obeys $p' = 2p^2 - p^4$. This map has two attracting fixed points, $p = 0$ (fragmented, $d_s \to 0$) and $p = 1$ (intact lattice, $d_s = 2$), and one repulsive fixed point $p_c = (\sqrt{5}-1)/2$, with $\nu = \ln 2/\ln(6-2\sqrt{5}) \approx 1.6353$. A non-integer spectral dimension therefore appears only at a tuned, unstable critical point. For the critical spanning cluster, including all dangling parts, the conditional moments of the cluster mass obey an exact 2×2 linear recursion. Its Perron eigenvalue $\Lambda = 7 - 2\sqrt{5} + \sqrt{73 - 32\sqrt{5}}$ gives the closed-form mass dimension $d_f = \log_2 \Lambda = 1.89929$. Population dynamics reproduces it to 0.1% after twenty levels. The resistance between the poles obeys a nonlinear recursion with Lyapunov exponent $\zeta = 0.8127 \pm 0.0002$, so the Einstein relation gives $d_s = 2d_f/(d_f + \zeta) \approx 1.401$. Exact diagonalisation of the graph Laplacian on generations 6 and 7, averaged over whole log-periods, gives 1.41 and 1.44. The same procedure gives 1.994 for the intact lattice, whose exact value is 2.

Away from p_c, $d_s(\tau)$ crosses over to 0 or 2 beyond $\xi \sim |p - p_c|^{-\nu}$. Newman–Watts shortcuts are a relevant perturbation that raises d_s beyond $\tau \sim \xi^2$ without creating a plateau. The DHL critical cluster has nearly the mass dimension of two-dimensional percolation, $91/48 = 1.8958$, but a very different resistance exponent (0.813 versus 0.983). Its spectral dimension, about 1.40, therefore differs from the $\mathbb{Z}^2$ value 1.317, from the mean-field value 4/3, and from the value 1.25 sometimes assumed. Geometric similarity does not imply a common transport universality class. The spectral dimension of "the critical cluster" also depends on the chosen subgraph and Laplacian, as the comparison with the Hambly–Kumagai diffusion shows.

Contents

1 Introduction **2**

2 Exact percolation renormalization on the diamond lattice **2**

3 Closed-form mass dimension of the critical cluster **4**

4 Resistance exponent and the critical spectral dimension **5**

5 RG stability and crossovers **6**

6 Transport universality **6**

7 Discussion and conclusion **7**

1 Introduction

Several approaches to quantum gravity find that the effective dimension of spacetime, measured by the spectral dimension of a diffusion process, depends on scale [1, 2]. Similar scale-dependent dimensions occur in disordered media and on fractals [3]. In earlier work we showed that the spectral dimension is an equipartition count, $d_s(\tau) = 2\tau\langle\lambda\rangle_\tau$, and that physical values must be read from scaling windows between cutoff effects [4]. Here we ask how d_s flows under an exact coarse-graining of a graph.

A tempting picture is that a discrete network, under a renormalization of its connectivity, undergoes a topological transition in which its spectral dimension "spontaneously jumps" from an integer to a stable non-integer value. A specific value such as $d_s \approx 1.25$ is sometimes posited for the resulting fractal, but we are not aware of a derivation of it. For comparison, generic branched polymers, the tree-like phase of two-dimensional quantum gravity, have $d_s = 4/3$ [13]. Three statements in this picture can be tested.

1. Is the non-integer fixed point attractive, i.e. reached spontaneously?
2. Is the change of d_s a jump?
3. What is the value of d_s at the critical point, and is it universal?

Hierarchical lattices make the RG exact [5, 6]. On the diamond hierarchical lattice the percolation transition and its scaling limit are understood rigorously [7]. Our answers are no; no, it is a crossover; and approximately 1.40 for the full critical cluster, which is not universal across lattices, subgraphs or Laplacians.

Main results.

- The mass dimension of the critical spanning cluster, including dangling parts, has the closed form $d_f = \log_2\bigl(7 - 2\sqrt{5} + \sqrt{73 - 32\sqrt{5}}\bigr) = 1.89929$ (Theorem 1).
- The resistance exponent is $\zeta = 0.8127 \pm 0.0002$. Together with the Einstein relation and exact diagonalisation, this gives $d_s \approx 1.40$–1.44 (Section 4).
- Off-critical and rewired graphs flow away from the critical value (Section 5).
- Two-dimensional percolation and the DHL share the mass dimension to 0.2%, but not the transport exponents (Section 6).

2 Exact percolation renormalization on the diamond lattice

The diamond hierarchical lattice D_n is defined recursively [5, 7]. D_0 is a single edge between two poles. D_{n+1} is obtained by replacing every edge of D_n by a diamond: two branches in parallel, each consisting of two edges in series. Equivalently, D_{n+1} consists of four copies of D_n arranged in a diamond. D_n has 4^n edges and $\frac{2}{3}(4^n + 2)$ vertices, and the graph distance between the poles is 2^n. The intact lattice has mass dimension 2. Its pole-to-pole resistance is independent of n, because two series edges in parallel with two series edges have resistance one. Its spectral dimension is 2, in the sense of eigenvalue counting [7].

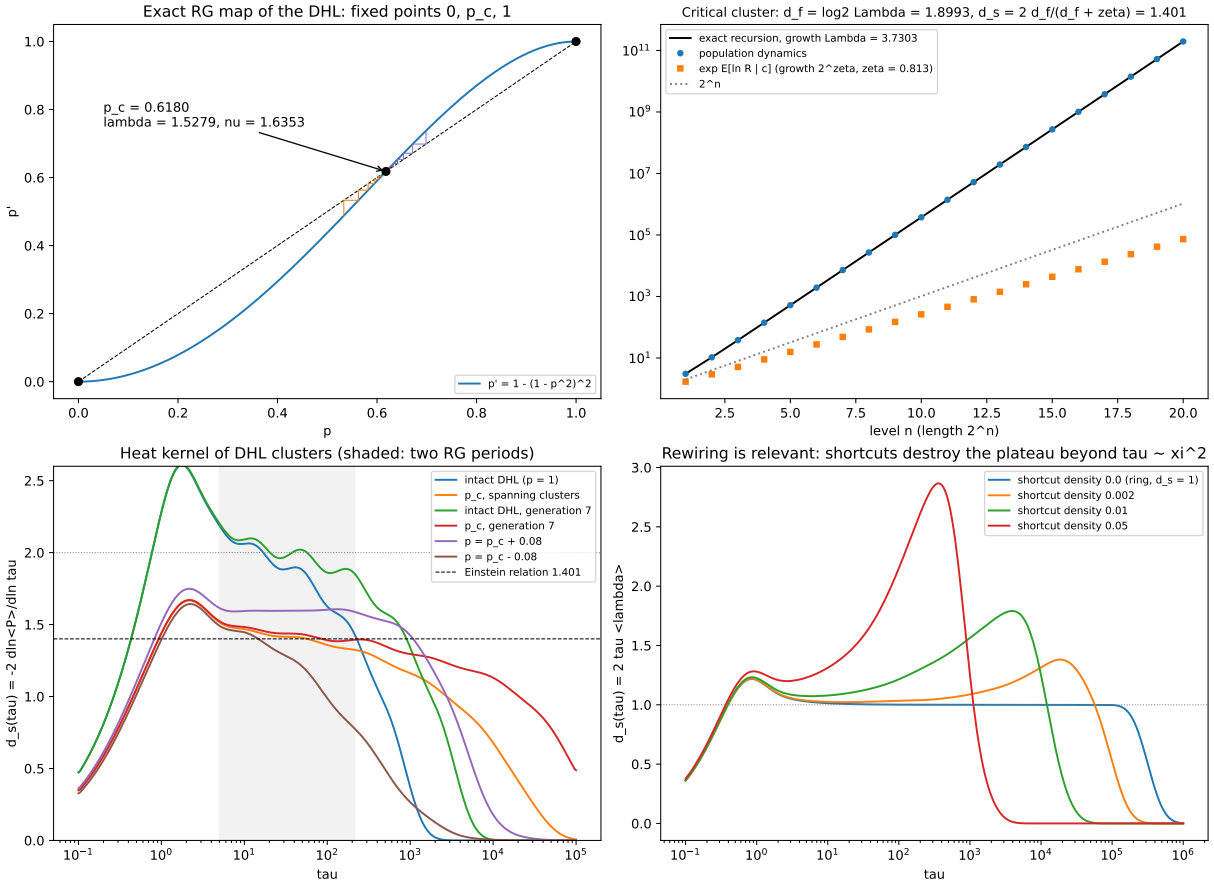


Figure 1: *Top left:* exact RG map $p' = 2p^2 - p^4$ with cobweb flows away from p_c. *Top right:* mean mass of the critical spanning cluster from the exact recursion (line) and from population dynamics (dots), and $\exp \mathbb{E}[\ln R \mid c]$ for the resistance between the poles (squares), which grows as $2^{\zeta n}$ with $\zeta = 0.813$, not as 2^n (dotted). *Bottom left:* $d_s(\tau) = -2\,\mathrm{d}\ln\langle P\rangle/\mathrm{d}\ln\tau$ from exact diagonalisation of the intact DHL and of percolation clusters at and near p_c; the shaded window covers two RG periods; the dashed line is the Einstein value 1.401. *Bottom right:* $d_s(\tau) = 2\tau\langle\lambda\rangle_\tau$ for a ring of 2000 sites with Newman–Watts shortcuts.

Declare each edge open independently with probability p, and let c_n be the event that the poles of D_n are joined by an open path. Since D_{n+1} is made of four independent copies of D_n, $P(c_{n+1}) = f(P(c_n))$ with

$$f(p) = 1 - (1-p^2)^2 = 2p^2 - p^4. \tag{1}$$

Proposition 2.1. *The map* (1) *has fixed points* 0, 1 *and* $p_c = (\sqrt{5}-1)/2$ *in* $[0,1]$. *The first two are attracting and* p_c *is repulsive, with*

$$\lambda = f'(p_c) = 4p_c(1-p_c^2) = 4p_c^2 = 6 - 2\sqrt{5} \approx 1.5279, \qquad \nu = \frac{\ln 2}{\ln \lambda} \approx 1.6353. \tag{2}$$

Proof. $f(p) = p$ reduces to $p(p-1)(p^2+p-1) = 0$. At p_c we have $p_c^2 = 1 - p_c$, hence $1 - p_c^2 = p_c$ and $f'(p_c) = 4p_c^2 = 4(1-p_c)$. $f'(0) = 0$ and $f'(1) = 0$. A deviation δp grows by the factor λ per level, i.e. per doubling of length, so $\xi \sim |\delta p|^{-\nu}$ with $\nu = \ln 2/\ln\lambda$. □

This is the standard percolation RG on the DHL [7]. Its message for the question of Section 1 is immediate. Every $p \neq p_c$ flows to 0 or 1, so a non-integer fixed point is reached only by tuning one parameter.

3 Closed-form mass dimension of the critical cluster

Let M_n be the number of open edges in the open cluster of pole 0 in D_n. By symmetry, the cluster of pole 1 has the same law. Put

$$P_n = P(c_n), \qquad a_n = \mathbb{E}[M_n \mathbf{1}_{c_n}], \qquad b_n = \mathbb{E}[M_n \mathbf{1}_{\neg c_n}], \tag{3}$$

with $(P_0, a_0, b_0) = (p, p, 0)$ for a single edge.

Lemma 3.1 (Series and parallel composition)**.** *Let X, Y be independent copies of a two-pole network with parameters (P, a, b).*

(i) Their series composition has $P_S = P^2$, $a_S = 2Pa$ and $b_S = (1-P)a + (1+P)b$.

(ii) The parallel composition of two independent networks with parameters (P_S, a_S, b_S) has $P_Q = 1-(1-P_S)^2$, $a_Q = 2a_S + 4P_S b_S$ and $b_Q = 2(1-P_S)b_S$.

Proof. (i) Let X join pole 0 to the middle vertex m, and Y join m to pole 1. The cluster of pole 0 consists of the cluster of pole 0 in X, together with the cluster of m in Y if X is crossed. So $M_S = M_X + \mathbf{1}_{c_X} M'_Y$, where M'_Y has the law of M by symmetry. Hence $\mathbb{E}[M_S \mathbf{1}_{c_S}] = \mathbb{E}[M_X \mathbf{1}_{c_X}]P + P\,\mathbb{E}[M'_Y \mathbf{1}_{c_Y}] = 2Pa$, and $\mathbb{E}M_S = (a+b)(1+P)$, which gives b_S.

(ii) If the parallel network is crossed, the poles lie in one cluster. A crossed branch contributes its pole cluster M_i. An uncrossed branch contributes both of its pole clusters, $M_i + M'_i$, because the other branch joins the poles. By independence of the branches, $a_Q = \sum_i \big(\mathbb{E}[M_i \mathbf{1}_{c_i}] + \mathbb{E}[(M_i + M'_i)\mathbf{1}_{\neg c_i}]\,P(c_j)\big) = 2a_S + 4P_S b_S$, where $j \neq i$. If it is not crossed, neither branch is, and $M_Q = M_1 + M_2$, so $b_Q = 2b_S(1-P_S)$. □

Theorem 1 (Mass dimension)**.** *At $p = p_c$ the vector (a_n, b_n) obeys $(a_{n+1}, b_{n+1})^{\mathsf{T}} = T\,(a_n, b_n)^{\mathsf{T}}$ with*

$$T = \begin{pmatrix} 12 - 4\sqrt{5} & 2\sqrt{5} - 2 \\ 2\sqrt{5} - 4 & 2 \end{pmatrix}. \tag{4}$$

Its Perron eigenvalue is

$$\Lambda = 7 - 2\sqrt{5} + \sqrt{73 - 32\sqrt{5}} = 3.730288562\ldots, \tag{5}$$

and the mean mass of the critical spanning cluster grows as $\mathbb{E}[M_n \mid c_n] = a_n/p_c \sim C\,\Lambda^n$. Since the pole distance is 2^n, the mass dimension is

$$d_f = \log_2 \Lambda = 1.8992872\ldots \tag{6}$$

Proof. Lemma 3.1 gives $a_{n+1} = [4P + 4P^2(1-P)]a_n + 4P^2(1+P)b_n$ and $b_{n+1} = 2(1-P^2)[(1-P)a_n + (1+P)b_n]$, with $P = P_n$. At $P = p_c$, $p_c^2 = 1 - p_c$ and $1 - p_c^2 = p_c$ give the entries (4). The characteristic polynomial is $x^2 - (14 - 4\sqrt{5})x + (4\sqrt{5} - 4)$, whose larger root is Λ. T has positive entries, and $(a_0, b_0) = (p_c, 0)$ has a positive component along the Perron vector. Hence $a_n \sim C\Lambda^n$ with $C > 0$, and $P_n = p_c$ for all n. □

Remark (The skeleton). Hambly and Kumagai describe the critical cluster through a multitype branching process. An edge is labelled c if the sub-diamond it represents is crossed [7]. Given that its parent is crossed, an edge is crossed with probability $p_c + (1-p_c)p_c^2$. The expected number of c-children of a c-edge is therefore $4(p_c + (1-p_c)p_c^2) = 12 - 4\sqrt{5}$, which is the entry T_{11}. The skeleton of edges that are c-labelled at every level thus has mass dimension $\log_2(12 - 4\sqrt{5}) \approx 1.6115$. It contains the current-carrying backbone and some, but not all, dangling edges.

Check. We simulate the recursion by population dynamics. At every level, the crossing indicator of each of the four sub-networks is drawn exactly with probability p_c, and its mass and resistance are drawn from a pool of 4×10^5 crossed or uncrossed samples of the previous level. Exact sampling of the indicator removes the drift that the repulsive fixed point would otherwise cause. After 20 levels the sampled mean mass is 1.9570×10^{11}, against the exact 1.9583×10^{11}. The last growth ratio is 3.7287, against $\Lambda = 3.7303$ (Figure 1, top right).

4 Resistance exponent and the critical spectral dimension

Resistance. Dangling parts carry no current, so the pole-to-pole resistance R_n of a crossed cluster satisfies the series and parallel laws. In series it is $R_X + R_Y$. In parallel it is $(R_1^{-1} + R_2^{-1})^{-1}$, with $R = \infty$ for an uncrossed branch. The recursion is nonlinear, and $R_n/2^n$ does not converge. We define the resistance exponent by the Lyapunov exponent

$$\zeta = \lim_{n\to\infty} \frac{1}{n}\, \mathbb{E}\big[\log_2 R_n \,\big|\, c_n\big]. \tag{7}$$

In the population dynamics the increments of $\mathbb{E}[\log_2 R_n \mid c_n]$ are stationary from about level 10. Over the last eight levels they give

$$\zeta = 0.8127 \pm 0.0002, \tag{8}$$

and the median of $R_n/2^n$ decreases geometrically ($0.128, 0.112, 0.099, 0.087, 0.076$ at levels 16–20). The reason is that, conditioned on a crossing, both branches are crossed with probability $P_S^2/(1-(1-P_S)^2) = p_c^4/p_c = p_c^3 = \sqrt{5} - 2 \approx 0.236$. At such a level the resistance stays roughly the same instead of doubling. Ignoring fluctuations this suggests $\zeta \approx 1 - 0.236$; the fluctuations of the branch resistances raise the value to 0.813.

Einstein relation. For a fractal with mass dimension d_f and resistance exponent ζ, the walk dimension is $d_w = d_f + \zeta$ and the spectral dimension is $d_s = 2d_f/d_w$ [8, 3]. Hence

$$d_s = \frac{2d_f}{d_f + \zeta} = \frac{2 \times 1.89929}{1.89929 + 0.8127} \approx 1.401. \tag{9}$$

This combines an annealed mass exponent with a typical resistance exponent. We therefore test it directly.

Exact diagonalisation. We sample bond configurations of D_6 (4096 edges) and D_7 (16384 edges) at p_c and keep the cluster of pole 0 when it reaches pole 1. For each sample we compute the full spectrum of the combinatorial graph Laplacian. From the annealed return probability $\langle P(\tau)\rangle = \langle V^{-1}\mathrm{Tr}\, e^{-\tau\Delta}\rangle$ we obtain $d_s(\tau) = -2\,\mathrm{d}\ln\langle P\rangle/\mathrm{d}\ln\tau$. For a single graph this is $2\tau\langle\lambda\rangle_\tau$ [4]. The curves oscillate log-periodically. One RG level multiplies time by $\Lambda 2^\zeta = 6.552$, so we average d_s uniformly in $\ln\tau$ over exactly two periods, $5 \le \tau \le 215$. The intact lattice, whose exact value is 2, calibrates the finite-size error. We also compute the quenched average, the mean over samples of $2\tau\langle\lambda\rangle_\tau$. Table 1 shows the results.

On D_7 the intact lattice is within 0.006 of its exact value, so the generation-7 estimate is reliable to a few hundredths. We conclude that the full critical spanning cluster has

$$d_s \approx 1.40\text{–}1.44. \tag{10}$$

The annealed and quenched averages agree to 0.002, so the choice of ensemble average is immaterial here. The residual difference from (9), which combines an annealed mass exponent with a typical resistance exponent, is of the size of the finite-size error on D_6 and the sampling error on D_7. The value 1.25 is excluded.

Table 1: Spectral dimension averaged uniformly in $\ln\tau$ over two RG periods ($5 \le \tau \le 215$), from the annealed return probability and as a quenched average over samples.

Graph	samples	mean size	annealed	quenched
intact DHL, D_6	1	2732	1.860	1.860
intact DHL, D_7	1	10924	1.994	1.994
critical spanning cluster, D_6	60	1791	1.412	1.414
critical spanning cluster, D_7	24	6567	1.438	1.439
Einstein relation (9)			1.401	

5 RG stability and crossovers

Off-critical flows. For $p = p_c \pm \delta$ the deviation grows by the factor λ per level. At scales below $\xi \sim \delta^{-\nu}$ the cluster looks critical, and above ξ it flows to the intact lattice ($d_s \to 2$) or to finite fragments ($d_s \to 0$). For $\delta = 0.08$, $\xi \approx 0.08^{-1.635} \approx 62$, which is comparable to the size $2^6 = 64$ of D_6. Accordingly, at $p_c - 0.08$ the curve $d_s(\tau)$ falls towards 0 inside the window. At $p_c + 0.08$ it stays near 1.60, between the critical and the intact values. At fixed τ, d_s changes continuously with p. Only the asymptotic exponent, 0, $d_s^{(c)}$ or 2, is discontinuous at p_c. The "jump" is a crossover.

Rewiring. The other perturbation proposed for geometric RG flows is rewiring. Add to a ring of $N = 2000$ sites a Poisson number, with mean ϕN, of random shortcuts (Newman–Watts model [9, 10]). The typical distance between shortcut ends is $\xi = 1/\phi$, which is relevant with $\nu = 1$ [10]. The ring has $d_s = 1$, with a small lattice overshoot to 1.218, the value $1.2178\,d$ found in [4]. With shortcuts, $d_s(\tau) = 2\tau\langle\lambda\rangle_\tau$ rises above 1 for $\tau \gtrsim \xi^2$. Its maximum is 1.38, 1.79 and 2.87 for $\phi = 0.002, 0.01, 0.05$ before finite-size collapse (Figure 1, bottom right). Rewiring therefore drives the graph away from any finite-dimensional plateau. It does not stabilise a non-integer value.

6 Transport universality

Table 2 collects the exponents of critical percolation clusters on several lattices.

Table 2: Mass dimension, resistance exponent and spectral dimension. Exponents are with respect to the Euclidean or hierarchical length; for the Hambly–Kumagai diffusion, θ is the dimension in its effective-resistance metric.

Object	d_f	ζ	d_s	Source
intact DHL	2	0	2	exact
DHL critical spanning cluster	1.89929	0.8127(2)	1.40–1.44	this work
DHL skeleton, renormalised form	$\theta = 5.2654$		$2\theta/(\theta+1) \approx 1.681$	[7]
$\mathbb{Z}^2$ critical percolation	91/48	0.9826(8)	1.317	[11, 12]
mean field ($d > 6$)	4	2	4/3	[8, 14]

Three conclusions follow.

1. *Geometry is not transport.* The DHL cluster reproduces the mass dimension of two-dimensional percolation to 0.2% (1.8993 against 1.8958). Its resistance exponent, however, is 0.813 against 0.983, and its spectral dimension is about 1.40 against 1.317. Migdal–Kadanoff-type hierarchical lattices approximate the geometric exponents of $\mathbb{Z}^2$, but they

are not in its transport universality class. Within the DHL the lower resistance exponent has a simple source (Section 4). Conditioned on a crossing, both branches of a diamond are crossed with probability $p_c^3 = \sqrt{5} - 2$ at every level, independently of scale, and such levels do not increase the resistance. On $\mathbb{Z}^2$, loops of all shapes and sizes contribute, and we do not claim a comparable mechanism there.

2. *"The" critical cluster is not unique.* Hambly and Kumagai construct a diffusion on the scaling limit of the c-labelled skeleton. Its Dirichlet form is renormalised so that the resistance across the lattice stays one, and its natural measure is the one carried by the branching process. In the resulting resistance metric they prove heat-kernel bounds with exponent $\theta/(\theta+1)$, $\theta = 5.2654$, i.e. spectral dimension ≈ 1.681 [7]. This is a different subgraph, measure and Laplacian from the full cluster with unit conductances. The difference of 1.68 and 1.40 shows that the spectral dimension depends on these choices, not only on the lattice.

3. *Mean field is not two-dimensional.* The Alexander–Orbach value 4/3 is proved in high dimensions [14]. Neither the $\mathbb{Z}^2$ value nor the DHL value equals it, and none equals 1.25.

7 Discussion and conclusion

In a model where the coarse-graining of a graph is exact, the spectral dimension behaves as RG theory predicts. Integer values (2, and 0 for fragments) are attracting fixed points. A non-integer value, $d_s \approx 1.40$ for the full DHL critical cluster, occurs only at the repulsive fixed point p_c and must be tuned. The change from integer to non-integer behaviour is a crossover at $\xi \sim |p - p_c|^{-\nu}$, not a jump. The critical value is determined by two exponents. The mass exponent has the closed form $\log_2\bigl(7 - 2\sqrt{5} + \sqrt{73 - 32\sqrt{5}}\bigr)$, and the resistance exponent 0.8127 is a Lyapunov exponent of a nonlinear recursion. Rewiring is a relevant perturbation that removes rather than creates non-integer plateaus.

This has two consequences for applications. In quantum gravity, a non-integer spectral dimension produced by a critical point of a geometric RG requires the tuning of a relevant coupling, as for the second-order transition lines of causal dynamical triangulations [1]. Its value depends on the transport properties of the critical geometry and on the Laplacian used, not only on its mass dimension. In biological growth, claims of a common "universality class" for fractal tissues or leaf margins would require an RG in which the relevant directions are identified and the transport exponents are measured. A fractal box-counting dimension alone does not determine the spectral dimension, as Table 2 shows. A stable non-integer d_s would need a mechanism that makes the critical point attractive, such as self-organised criticality. This is a hypothesis to be tested, not a consequence of coarse-graining.

Data and code availability

The script `rg_spectral_dimension.py`, the data (CSV), the figure and the console output are available at `https://github.com/Ruqing1963/rg-spectral-dimension` and archived under DOI 10.5281/zenodo.23201028.

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